Speaker notes: Name That LLM

Read these while showing `documents/presentation.pdf`, one block per slide. About 480 words, so three minutes at a brisk, steady pace. Say “glm-5.3” as “G L M five point three”.

1. Cover (0:00)

> Hi, I’m Nabil. This is my solo entry to Name That LLM.

2. The task (0:04)

> Each passage was written by one of eighteen models. I train on 2,699 passages and am scored on about four thousand, on prompts I never see. Labs are easy to tell apart. Siblings are not: the two GLM models, and Gemini 3.7 against 3.8.

3. Literature review (0:17)

> I wrote no code for the first two hours. I read twenty-two public notebooks and eighteen discussion threads, and wrote fifteen research notes, each ending in a decision. Three findings set the plan: data beats model size, the best blend’s weights were public, and no public model had honest out-of-fold scores, so I had to validate on my own folds.

4. Starting point (0:37)

> So I froze the folds first, grouped by prompt, like the hidden test. A plain n-gram SVC scored 0.348. Then I wrote my own loader for the published 0.620 weights and built on top.

5. Approach (0:50)

> Each member gives log-probabilities. They are blended, a few extra members join at fixed shares, and half-strength balancing evens out the classes. The filled boxes are mine: the SVC, a Gaussian group head and my own Qwen3-8B LoRA, trained overnight on Kaggle.

6. Hypotheses (1:06)

> Four ideas. Style lives in small habits, so n-grams should work. Data beats size: public passages took the SVC to 0.484. The prompt hides the author: take out what passages on one prompt share, and style is left. And the test mix is uneven, so balance only halfway.

7. Ablation, leaderboard (1:24)

> One change per probe. The spine with my SVC: 0.626. Half-strength balancing: 0.629. Two public seeds: 0.631. Then four ties at 0.631. Probe nine, the Gaussian head at full weight plus ModernBERT, reached 0.637. My picks are probe nine, and probe seven, which carries my own model.

8. Ablation, cross-validation (1:45)

> These scores come from folds each model never saw. My own LoRA reached 0.519, level with a 27-billion model. The Gaussian group head added 0.058, my biggest single gain. Other ideas stayed within noise.

9. What didn’t work (2:00)

> Peer correction removes a topic’s shared lean and drops class quotas. It helped its author’s weaker blend. On mine it fell to 0.622: without quotas, a strong blend drifts towards the models it already favours.

10. What the leaderboard told me (2:14)

> Small gains round away on the public slice, so I steered by cross-validation. And features must be honest: the head helped on my own model, but on the 14B models, which saw all of train, it dropped to 0.630.

11. How I used AI (2:27)

> I used Claude Code from the first hour, set up like my other repos: an AGENTS file with the rules as invariants, small tickets, a research note before each decision and a log row for every change. Frozen folds, tests first and every model on Kaggle make it reproducible. That structure let six side sessions work in parallel while I made every call.

12. What I'd try next (2:47)

> Next: a bigger open pool, five folds per member for honest blending, joint decoding across members, and targeted data for the sibling pairs. Thanks for watching.